



Maternal exposure to different sizes of polystyrene microplastics during gestation causes metabolic disorders in their offspring[☆]

Ting Luo, Yi Zhang, Caiyun Wang, Xiaoyu Wang, Jiajie Zhou, Manlu Shen, Yao Zhao, Zhengwei Fu, Yuanxiang Jin^{*}

College of Biotechnology and Bioengineering, Zhejiang University of Technology, Hangzhou, 310032, China

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ABSTRACT

Microplastics (MPs) are highly concerned environmental pollutants that are ubiquitous in the environmental and might affect human and animal health. In this study, we exposed pregnant mice to 0.5 and 5 μm with 100 and 1000 $\mu\text{g/L}$ polystyrene MPs, then investigated maternal MPs exposure during gestation and evaluated the potential effects on the mice offspring (PND42). In the F1 offspring, the serum triglyceride (TG), total cholesterol (TC), high-density lipoprotein cholesterol (HDL-C) and low-density lipoprotein cholesterol (LDL-C) levels and hepatic TC, TG levels were altered, while some of them were only significant in 5 μm MPs-treated group. Various serum metabolites including amino acids and acyl-carnitines were carried out by nonderivatized tandem mass spectrometry, there were 11 and 15 kinds of metabolites changes significantly in 0.5 and 5 μm MPs-treated groups, respectively. Furthermore, the changes of C0 and C0/(C16 + 18) indicators suggested the potential risk of fatty acid metabolism disorder, which was verified by hepatic genes expression. These results indicated that maternal exposure of two different sizes of polystyrene MPs increased risks of metabolic disorder in their offspring, and greater effects were observed in 5 μm MPs-treated groups. The data provides a preliminary exploration of the potential relationship between MPs and the risk metabolic disorder even in the next generation, which might offer new insights into the health risk assessment of MPs.

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1. Introduction

"Microplastics" (MPs) was first used by Richard C. Thompson to describe small plastics particles (Thompson et al., 2004), nowadays MPs usually refers to plastic particles with a particle size of less than 5 mm (Moore, 2008). It is estimated that there are more than 5 trillion pieces of plastic debris floating in the oceans around the world, and the weight is more than 250,000 tons, which will bring harmful effects on aquatic ecosystems (Eriksen et al., 2014; Chae and An, 2017). Moreover, MPs also can be transported over long distances by external forces such as wind, rivers, and ocean currents, consequently, the MPs are widely distributed environmental media such as global waters, sediments, soils and even atmosphere (Barnes et al., 2009; Cole et al., 2011; Browne et al., 2011; Bergmann

et al., 2017; Abbasi et al., 2019). Hence, MPs pollution has gradually developed into a global environmental problem, and received attention of various areas including government, society and academia. It is a research hotspot in the field of international ecology and environmental science.

According to previous search, particle abundances of over 1×10^4 particles m^{-3} have been reported for the surface water of China's freshwater ecosystems (Zhao et al., 2014; Di and Wang, 2018; Yan et al., 2019). The concentration of MPs has even reached up to 2.58×10^4 particles m^{-3} in Taihu Lake and approximately 1×10^6 particles m^{-3} in Los Angeles River or much more in some heavily contaminated areas (Su et al., 2016; Eerkes-Medrano et al., 2015). Sludge disposal such as land application is suspected as a significant source of MPs pollution and the MPs were observed in some areas of China ranged from $1.60\text{--}56.4 \times 10^3$ particles per kilogram of dry sludge (Li et al., 2018). Due to its small size, MPs are easily mistaken as food and ingested by aquatic organisms (Moore et al., 2001; Moore, 2008). A number of studies proved that MPs can pass along the food chain and even enriched at higher trophic levels (Farrell and Nelson, 2013; Setälä et al., 2014), then eventually enter

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^{*} Corresponding author. 18, Chaowang Road, College of Biotechnology and Bioengineering, Zhejiang University of Technology, Hangzhou, China. Tel./fax: +86 571 88320742.

E-mail address: jinyx@zjut.edu.cn (Y. Jin).

the human body. Seafood consumption is clearly one of the possible pathways for human MPs exposure. As reported in previous studies, MPs were widely distributed in shellfish and fish, for example, *Alectryonella plicatula* (10.78 ± 4.07 particles/individual), *Patinopecten yessoensis* (57.17 ± 17.34 particles/individual), *Alepes djedaba* (8.00 ± 1.22 item/10 g fish muscle), *Epinephelus coioides* (7.75 ± 2.16 item/10 g fish muscle), etc (Li et al., 2015; Barboza et al., 2018; Akhbarizadeh et al., 2018). In addition, MPs were also found in bottled water of different countries, some commercial sea salt and even some canned food (Yang et al., 2015; Karami et al., 2017; Mason et al., 2018; Karami et al., 2018). The consumption of these products is extremely common in life, which provide much more potential pathways for MPs to enter the human body and then increase the health risk.

A recent study was presented in Vienna at UEG Week (Schwabl et al., 2018), reporting that MPs have been found in human stools from different countries for the first time. There were up to nine different plastics were found and the particle sizes of them were ranging from 50 to 500 μm . This report suggested that tiny particles may be widely present in the human food chain (Amanda, 2018), strongly reminded us again that we should pay more attention to the potential impacts of MPs on health.

According to previous studies, MPs $\leq 20 \mu\text{m}$ can enter the organ, and the smaller MPs with 0.1–10 μm particle size might can enter all organs of animals, pass through the cell membrane, blood-brain barrier and placenta, and even transfer to the next generation (Browne et al., 2008; von Moos et al., 2012; Bouwmeester et al., 2015; Wright and Kelly, 2017). Actually, it was proved that polystyrene MPs (5 and 20 μm) can accumulate in tissues including liver, kidney and gut, and induce oxidative stress and metabolic changes (Deng et al., 2017; Deng et al., 2018). Subsequently, our previous works evaluated the effects of polystyrene MPs with different particle sizes on mice and indicated that the polystyrene MPs can induce gut microbiota dysbiosis, gut barrier dysfunction and affect liver lipid metabolism in male mice (Lu et al., 2018; Jin et al., 2019).

However, data on animal health risk assessment of MPs is still very limited, not to mention intergenerational effects, especially for more advanced rodents and even humans. Here, we established a mouse model involving maternal polystyrene MPs exposure during gestation (pregnant mice exposure to 0.5 and 5 μm with 100 and 1000 $\mu\text{g/L}$ polystyrene MPs in drinking water) to evaluate intergenerational effects on their offspring caused by MPs. The concentrations of MPs were used in our previous studies (Lu et al., 2018; Jin et al., 2019), which caused some adverse effects in male mice. Here, in order to better explore the intergenerational effects of MPs, the concentrations of MPs (100 and 1000 $\mu\text{g/L}$) chosen were higher than those in the normal environment, but much lower than those in some previous studies. The results indicated that maternal MPs exposure could alter the metabolite levels and increase the risks of fatty acids metabolism disorder in the offspring, and the effects may be related to MPs particle sizes.

2. Materials and methods

2.1. Chemicals

Pristine polystyrene MPs (with the particle sizes 0.5 and 5 μm in diameter) were purchased from Microspheres-Nanospheres (New York, USA) as forms of suspension. Both sizes of polystyrene MPs were checked in the previous studies (Jin et al., 2018; Wan et al., 2019). The polystyrene MPs was used as received (polystyrene MPs suspension) and the stock solution was treated with ultrasound at least for 30 min before dilution.

2.2. Animal husbandry

7-week-old mice (ICR) were purchased from the China National Laboratory Animal Resource Center (Shanghai, China). All the mice were kept in the animal facilities, individually housed, (strip lights were used to illuminate, 200 lux at cage level with a photoperiod of 12 h light to 12 h dark; $22 \pm 2^\circ\text{C}$). After a week of accommodation, the female mice (defined as F0) individually caged with age-matched males for fertility.

After copulation, F0 female mice were separated after positive identification of a vaginal sperm plug, and this day was defined as gestational day zero (GD 0). Then the pregnant mice were randomly divided into five groups to ensure that there were no significant differences in average body weights among each group at the beginning of the experiment ($n = 5-6$ in each group). In this study, control mice received ordinary water, while the experimental female mice received drinking water with the 0.5 μm -100 $\mu\text{g/L}$ (0.5 μm -L), 0.5 μm -1000 $\mu\text{g/L}$ (0.5 μm -H), 5 μm - 100 $\mu\text{g/L}$ (5 μm -L) and 5 μm - 1000 $\mu\text{g/L}$ (5 μm -H) polystyrene MPs during gestation (from GD 0 to production day).

The day of parturition was usually defined as postnatal day zero (PND 0). After weaning, F0 female were killed while the F1 female and male offspring were separated and housed 3 per cage, respectively. Mice offspring from each group received ordinary drinking water and standard rodent chow ad libitum. At PND 42, the F1 generation female and male offspring from each group were killed. All mice were fasted for approximately 12 h and anesthetized by ether before being sacrificed. The serum and tissue samples were collected for further use, which were immediately stored at -80°C as quickly as possible. The experimental designs are shown as Fig. 1. The Guiding Principles for the Use of Animals in Zhejiang University of Technology was strictly complied in this study. At the same time, every effort was carried out minimize the pain of experimental animals.

2.3. Determination of serum and hepatic parameters

The serum total cholesterol (TC), triglyceride (TG), high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C) levels and hepatic TC, TG levels were measured according the protocols of corresponding kits, respectively. These kits were provided by Nanjing Jiancheng Institute of Biotechnology (Nanjing, China). In addition, hepatic TC, TG levels were normalized by the total protein contents of their hepatic homogenates. Total protein contents of liver homogenate were determined using the commercial BCA Protein Assay Kit, the kit was purchased from Beyotime Institute of Biotechnology (Shanghai, China). The serum NEFA levels were determined using LabAssay™ NEFA kits from Your FUJIFILM Pure Chemical Corporation (Chou-Ku, Japan) according to their instructions.

2.4. Total RNA extraction and gene expression analysis

The TRIzol Reagent (Takara Biochemicals, China) was used for total RNA extraction of livers. Then cDNA was synthesized using a reverse transcriptase kit (Vazyme, China) according to their instructions. The obtained cDNA was used in an RT-qPCR experiment to evaluate the expression level of the relevant genes. RT-qPCR was carried out using the SYBR Green system (Vazyme, China) in an Eppendorf MasterCycler ep RealPlex² (Wesseling-Berzdorf, Germany). The transcription level of the 18S rRNA gene was measured as a housekeeping gene, and the transcription of target genes was normalized to the corresponding 18S transcription level in each sample. The sequences of the primer were in some previous study and show as Table S1 (Wu et al., 2018; Jin et al., 2019; Wang et al.,

Experimental design

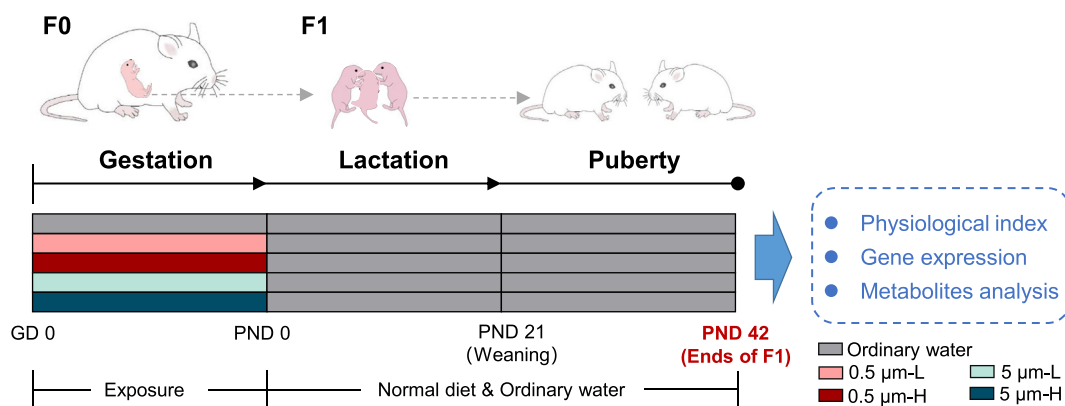


Fig. 1. Diagram of experimental design. Pregnant ICR female mice received polystyrene microplastics (MPs) with particle sizes of 5 µm and 0.5 µm (100 µg/L and 1000 µg/L in drinking water) during gestation, while control mice received ordinary water. F1 offspring were killed at PND 42.

2019a). In this study, RT-qPCR procedure is as follows: denaturation for 1 min at 95 °C, followed by 40 cycles of 15 s at 95 °C and 1 min at 60 °C. The relative quantification of expression level of target genes in MPs-treated groups was performed refer to the methods in previous study (Livak and Schmittgen, 2001; Wang et al., 2019b).

2.5. Serum sample preparation and nonderivatized tandem mass spectrometry

Firstly, fresh serum samples ($n = 5$ in each group) from F1 female and male mice were randomly selected, and 30 µL serum samples were dropped into each circle of filter paper blood cards (BM2010-4, Hangzhou, China) uniformly and respectively. To avoid stacking, we guaranteed that only one drop of serum was used at a time. Then, the filter paper blood cards containing the serum samples were placed in a cool, ventilated environment, dried at room temperature (20–25 °C) for at least 3 h.

The measurement of amino acids, carnitine and succinylacetone with the NeoBase assay involves extraction of NeoBase Non-derivatized MS/MS Kits dried sample spots from each group with a solution containing stable-isotope labeled internal standards and analysis using a tandem mass spectrometry (MS/MS) system. The response of each analyte relative to their corresponding stable-isotope labeled internal standard was proportional to analyte concentration. According to the instruction of NeoBase Non-derivatized MS/MS Kits, data was acquired in multiple reaction monitoring (MRM) mode. During this acquisition, a collisionally induced product of each analyte is measured for a set time period. Data collection and processing was performed by software packages within the system. Further analysis of the detected serum metabolites was performed using SIMCA 14.1 (Umetrics, Sweden).

2.6. Statistical analysis

In the present study, differences among the data from each group were evaluated by a one-way ANOVA followed by Dunnett's protected least significant difference test using SPSS 23.0 (SPSS, Chicago, Illinois, USA). All the data from each MPs-treated group were compared with the control group, and the values are expressed as the means $\pm$ SEM (Standard Error of Mean). Differences with $p < 0.05$ was accepted as significant and $p < 0.01$ was considered as very significant.

3. Results

3.1. Effects of maternal MPs exposure on F1 growth phenotypes

The survival rate and sex ratio of the mice offspring were calculated and no obvious changes were observed in all MPs treated groups (Fig. S1), indicating that the maternal MPs exposure did not affect the survival rate and sex ratio of their offspring.

At PND 21, F1 female and male offspring were separated into different cages. Body weights of F1 female and male were assessed, no significant statistical differences were observed (Fig. 2). However, through the average weights of each group, the offspring from MPs-treated groups appeared to be more inclined to have a smaller body weight. In addition, the relative liver weights were not changed significantly in each 0.5 µm MPs-treated group, while this index showed an upward trend in 5 µm MPs-treated groups of both F1 female and male (Fig. 2A and B).

3.2. Effects of maternal MPs exposure on biochemical parameters in the serum and liver

Maternal MPs exposure altered the biochemical parameters of serum and liver (Fig. 3). Serum TC levels of female showed a downward trend, but no statistical significance was observed; the serum TC levels of male increased significantly in each MPs-treated group. The serum TG levels of female in 0.5 µm and 5 µm MPs-treated group were lower compared with control while the statistical significance was just observed in 5 µm-L and 5 µm-H. For the male mice, the serum TG levels in each test group were significantly lower than those in the control. Serum HDL-C and LDL were also evaluated, they were no significant changes in female offspring. Nevertheless, HDL-C levels decreased for male in 0.5 µm-L, 0.5 µm-H and 5 µm-H, LDL-C levels significantly elevated compared with the male control (Fig. 3A).

Subsequently, biochemical parameters in the liver were also evaluated. The results showed that hepatic TC levels were lower than control and significant in 5 µm-L and 5 µm-H for both F1 female and male. In addition, the hepatic TG levels had analogous decreasing trends as the serum TG in each group whereas were not significant (Fig. 3B).

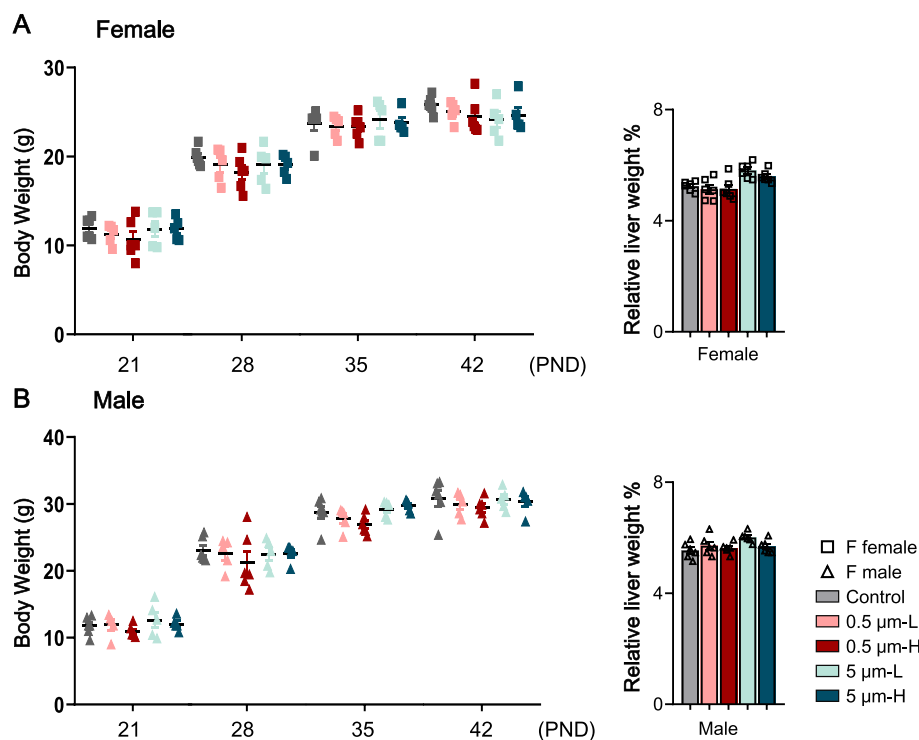


Fig. 2. Effects of maternal MPs exposure the growth phenotypes of F1 offspring. (A) Average body weights and relative liver weights of female. (B) Average body weights and relative liver weights of male. The presented values are the means $\pm$ SEM ($n = 5-6$). *, $p < 0.05$, **, $p < 0.01$ versus control.

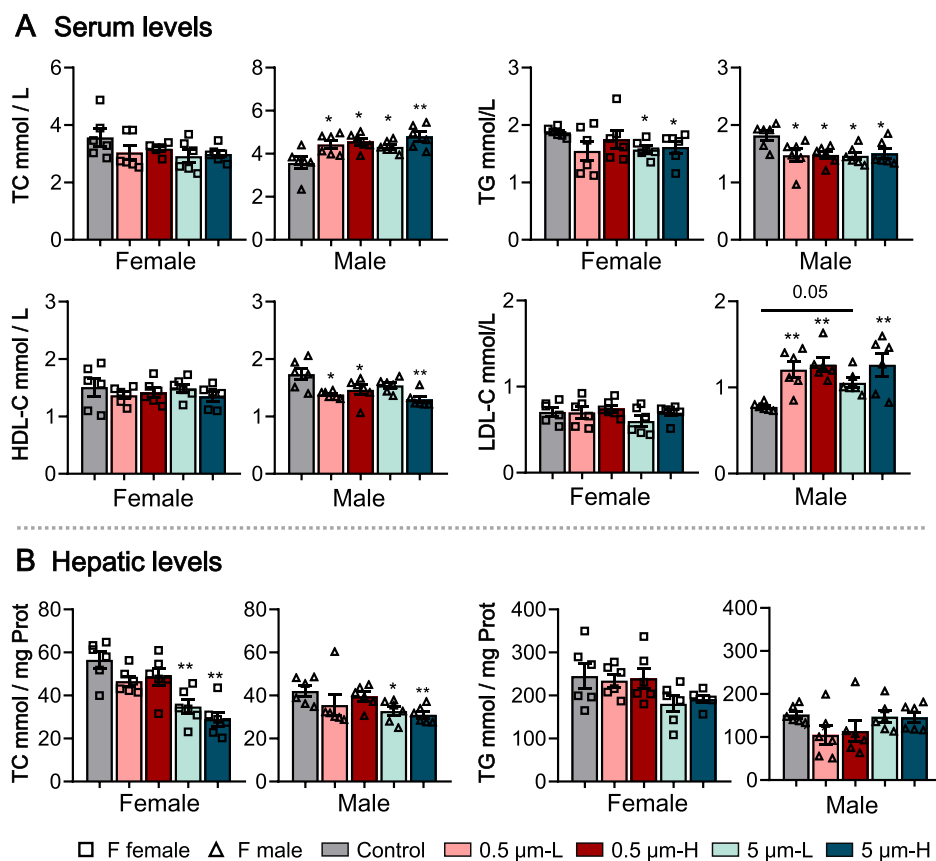


Fig. 3. Effects of maternal MPs exposure on biochemical parameters. (A) Serum TC, TG, HDL-C and LDL-C levels. (B) Hepatic TC, TG levels. The presented values are the means $\pm$ SEM ($n = 6$). *, $p < 0.05$; **, $p < 0.01$ versus control.

3.3. Effects of maternal MPs exposure on serum metabolites in the F1 generations

We further analyzed the serum metabolites including the amino acids, succinylacetone and carnitines in serum of F1. The serum metabolites were measured by nonderivatized tandem mass spectrometry. Through our results, we found that in the F1 female mice, the contents of arginine (ARG) trended to decrease and statistical significance was observed in 5 μm -H; the Leucine/isoleucine/hydroxyproline (LEU, etc.) levels in 0.5 μm -H and 5 μm -L group were much lower than control. Simultaneously, methionine (MET) and valine (VAL) contents were also much lower than those in the F1 female control. Overall, for female offspring, most amino acids also showed downtrends after maternal MPs exposure. In addition to this, succinylacetone (SA) levels were not affected in both F1 female and male (Fig. 4A). The contents of free carnitine (C0) were observed much higher in each MPs-treated group, while the contents of acetylcarnitine (C2) tended to increase and significance was observed in 5 μm -H group for female offspring (Fig. 4B).

As for male offspring, five kinds of amino acids were altered after maternal MPs treatment. The contents of alanine (ALA) increased significantly in 0.5 μm -L group and 5 μm MP treated groups. The contents of citrulline (CIT) and tyrosine (TYR) were much higher in 0.5 μm -H and 5 μm -H when compared with male control. In addition, the levels of methionine (MET) were significantly increased in most of these MPs-treated groups while the contents of proline (PRO) had a similar increase trend and just significant in 5 μm -H (Fig. 4A). Interestingly, for male offspring, most amino acids contents displayed increasing trends in each test group. The contents of the free carnitine (C0) were not changed, while a number of acyl-carnitines tended to decrease after maternal MPs treatment for male offspring. For instance, the contents of methylmalonic/3-hydroxy-isovaleryl carnitine (C3DC/C4OH) significantly lower in 5 μm -H group and the contents of glutaryl carnitine/3-hydroxy-hexanoyl carnitine (C5DC/C6OH) decreased significantly in three MPs-treated groups. In addition, many long chain acylcarnitines were also affected, hexadecenoyl carnitine (palmitoyl carnitine, C16), hexadecenoyl carnitine (C16:1) and octadecadienoyl carnitine (stearoyl carnitine, C18), octadecadienoyl carnitine (oleyl carnitine, C18:1) and octadecadienoyl carnitine (linoleyl carnitine, C18:2) levels were also much lower after maternal MPs treatment especially in 5 μm MPs-treated groups for male offspring (Fig. 4B).

Detailed changes of all significantly changed serum metabolites are shown as Fig. S2. According to the statistics, there were 11 kinds of serum metabolites altered by 0.5 μm MP, while 15 kinds of them significantly changed after 5 μm MP maternal treatment.

3.4. Further analysis and screening of the important serum metabolites in offspring

Further analysis of serum metabolites was carried out by SIMCA 14.1 (Umetrics, Sweden). The aim of the analysis was to screen for potential representative metabolites, which might be associated with the occurrence of metabolic diseases resulting from maternal MPs exposure. Firstly, all data were set into four pairwise comparison groups: female/male 0.5 μm MPs-treated group versus their control, female/male 5 μm MPs-treated group versus their control. The score plots of OPLS-DA showed good separation in all treatment groups (Fig. 4C, D, E, F). Our results indicated that serum metabolic profiles were disturbed following maternal MPs exposure.

Each point in the S-plots diagram represents a kind of metabolite, and the distribution of the variables indicates the trend of the related differential metabolites, the farther distance is, the greater

its contribution. Through the S-plots of female 0.5 μm MPs-treated group versus control, the farthest point was glycine (GLY) which was not statistically significant in the previous analysis (Fig. 4C), thence the second point representing free carnitine (C0) was chosen. On the other side, methionine (MET) might be the other potential differential metabolite. Similar analysis was carried for other pairwise comparison groups. Free carnitine (C0) and methionine (MET) were filtered out between female control and 5 μm -MPs; alanine (ALA) was the potential differential metabolites for male 0.5 μm MP and 5 μm MPs-treated groups (Fig. 4D, E, F).

In addition to this, comprehensive analysis of all the female through OPLS-DA model indicates the difference in metabolism between female and male (Fig. S3A). The score plots of OPLS-DA of female/male MPs-treated groups versus their control display similar results as above (Figs. S3B and C; Fig. 4). Taken together, maternal MPs exposure could disturb amino acid metabolism, carnitine metabolism which might have the important relationship with C0 and ALA.

3.5. Maternal MPs exposure caused fatty acid metabolic disorder in the F1 offspring

The levels of C0 and C0/(C16 + 18) are important indicators for clinical screening of genetic diseases (Wilcken and Wiley, 2008; El-Gharbawy and Vockley, 2018). For female F1 offspring, serum C0 levels were much higher than control after maternal MPs treatment; for male F1 offspring, just a slight upward trend (Fig. 5A) was observed. The serum C0/(C16 + C18) ratios showed similar increasing trend and significant in male (Fig. 5B). These findings suggested that maternal MPs exposure might lead to disturbances in carnitine metabolism and affect fatty acid (FA) metabolism.

To determine whether maternal MPs exposure would affect the fatty acid metabolism, serum NEFA and hepatic genes related to fatty acid metabolism were further analyzed. The serum NEFA levels tended to decline for female offspring, whereas, for male offspring, those were lower in 0.5 μm -L and 5 μm -L group (0.5 μm -H with $p = 0.06$, 5 μm -H with $p = 0.07$) compared with the control (Fig. 5C). Subsequently, hepatic transcriptional levels of related genes were detected through RT-qPCR. According to the results, the relative expression of genes regulating fatty acid transportation (*Fabp1*, *Fatp2*), β -oxidation (*Ppar- α* , *Acox*, *Cpt1- α* , *Mcad*) and fatty acid synthesis (*Srebp1c*, *Fas*, *Acl*, *Scd1*) significantly decreased in MPs-treated groups for both female and male offspring. The transcription levels of *Fat* increased significantly in 5 μm MPs-treated groups for female and 0.5 μm MPs-treated groups for male. As for expression of *Ppar- γ* , increased in 5 μm -H for female while not changed for male (Fig. 5D and E). In brief, maternal MPs exposure indeed induced the fatty acid metabolism disorder, and the effects of 5 μm MP seems more obvious.

4. Discussion

A lot of previous studies had proved that MPs could do harm to various organism, its toxicological effects were mainly reflected in growth and development toxicity, behavioral toxicity, immunotoxicity, reproductive toxicity and genotoxicity et al. (Browne et al., 2008; Sussarellu et al., 2016; Chae and An, 2017; Lu et al., 2019). In general, most of the subjects in these previous researches were aquatic organisms, while the understanding of the impacts of MPs on advanced land animals is limited. In the present study, a mouse model involving maternal polystyrene MPs exposure was used, the results of our research demonstrated that maternal MPs treatment during gestation could bring a series of intergenerational effects on their offspring, and might relevant to the MPs particle sizes.

According to our results, average survival rates and sex ratio of

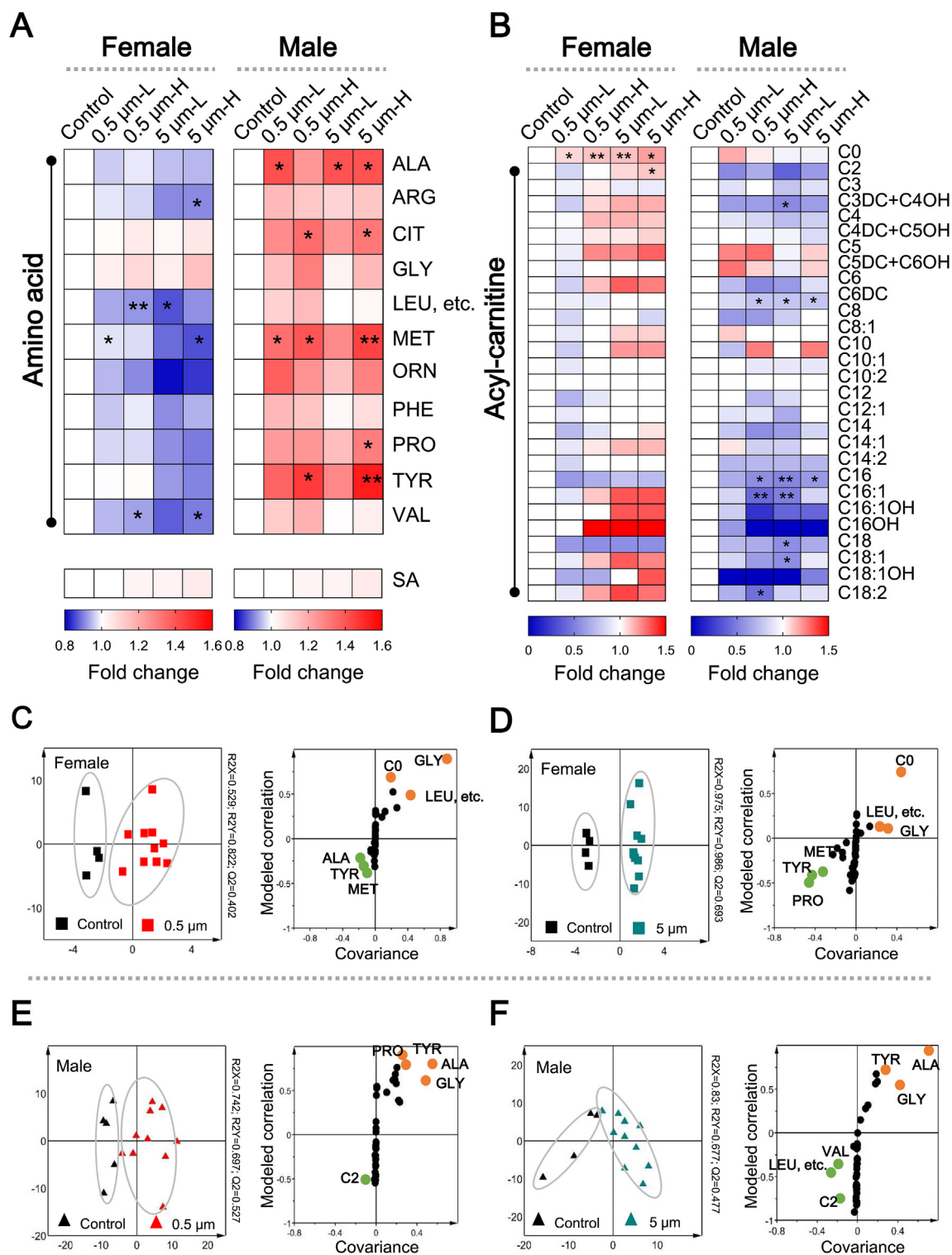


Fig. 4. Effects of maternal MPs exposure on the serum metabolite changes in the amino acids succinyl acetone and carnitine. (A) Heat map of various amino acids and succinyl acetone. (B) Heat map of free carnitine and acyl-carnitines. The data in the heat map were fold changes between MPs-treated group and control group, and metabolites with a value of 0 were excluded. In brief, red represents an upward trend and blue represents a downward trend. The full names of serum metabolites are shown as Table S2. OPLS-DA scored plots and corresponding S-plots were obtained from pairwise comparisons. (C) female 0.5 μm MPs-treated group versus female control; (D) female 5 μm MPs-treated group versus female control; (E) male 0.5 μm MPs-treated group versus male control and (F) male 5 μm MPs-treated group versus male control. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

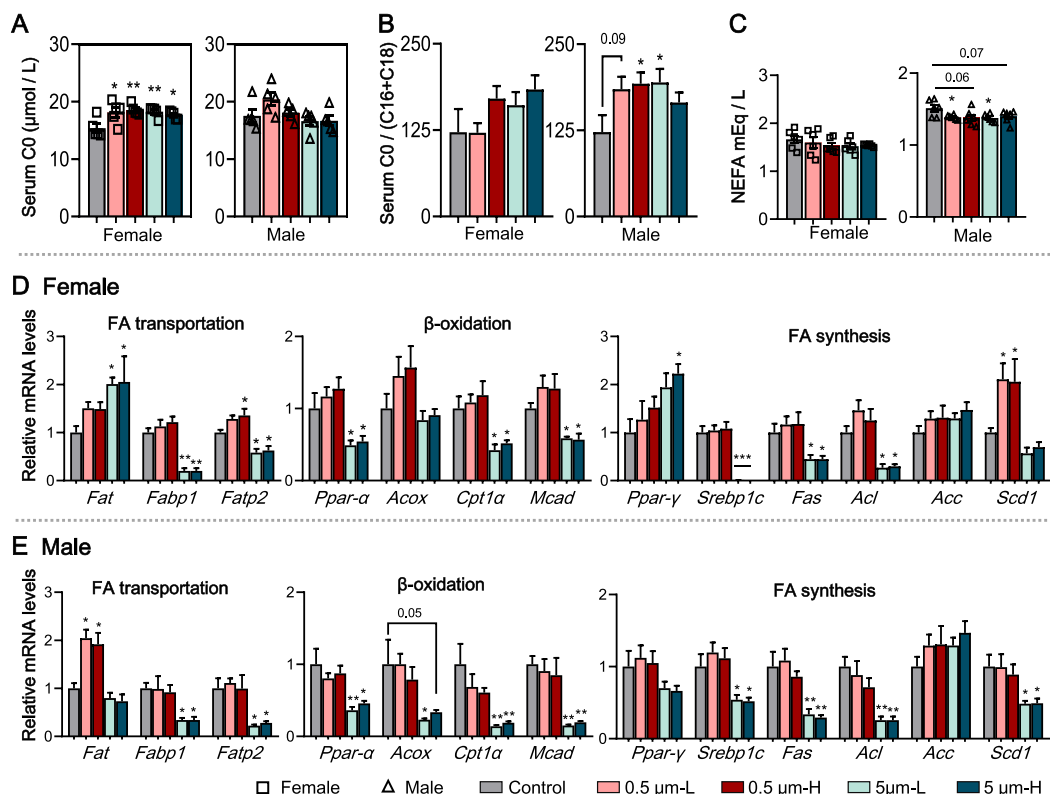


Fig. 5. Maternal MPs exposure induced hepatic fatty acid metabolic disorder. (A) Serum C0 levels of female and male; (B) Serum C0/(C16 + C18). (C) Serum NEFA levels of female and male. Gene expression in liver related to FA transportation, β -oxidation and FA synthesis of female (D) and male (E). The presented values are the means $\pm$ SEM (n = 6). *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$ versus control.

offspring were not affected, and average body weights seem to be showing a downward trend while no statistically significant were observed (Figs. 2 and S1). Recently, Lo and Chan (2018) reported that at environmentally relevant concentration (ten 2-mm MPs beads/mL), larval and juvenile *C. onyx* was not affected by MPs. At higher concentrations (30, 70 MPs beads/mL), these MPs treated larvae consumed a similar amount of algae but grew relatively slower compared to those in control. A similar result was carried out in mice experiment, average body weights tended to decrease when the mice were exposed to a high dose of 0.5 and 50 μ m (1000 μ g/L) polystyrene MPs for three and four weeks, respectively (Lu et al., 2018). In contrast, some studies also noticed that plastics might or might not affect growth. For example, Bruck and Ford (2018) found that ingesting low concentrations of 8 μ m MP (about 0.9, 9 and 99 MPs/g) did not impair the feeding or growth of amphipods along the exposure period, suggesting that the concentration of the MPs is an important factor for its adverse effects.

Here, the biochemical parameters involving serum TC, TG, HDL-C and LDL-C levels and hepatic TC, TG levels were altered at different degrees (Fig. 3). The serum TG levels and hepatic TC, TG levels of F1 male offspring displayed a particle size dependent effect while the statistical significance was observed only in 5 μ m MP-treated groups (Fig. 3A). Interestingly, Deng et al. (2017) reported that the TC and TG levels of livers were significantly decreased in male mice after 4 weeks exposure to 5 and 20 μ m MP with the concentrations of 0.01, 0.1 and 0.5 mg/day. Simultaneously, the changes of biological parameters related to energy metabolism and lipid metabolism including ATP, LDH, TC and TG levels showed good concentration dependence in this study. In another research (Lu et al., 2018), male mice were exposed to two particle sizes of 0.5 and 50 μ m MP for 5 weeks, hepatic TC and TG

levels have similar downtrends as those in this study. And the statistical significance was observed in the treatment group with the high MPs concentration 1000 μ g/L. It's worth mentioning that, the male mice were directly exposed to MPs in those two studies above so that displayed a good concentration dependent effect. According to the results of biochemical parameters, the concentration dependent effect was not obvious in the present study which might result from the complexity of intergenerational effects. In brief, MPs, like other environmental pollutants including heavy metals, fungicides and perfluorinated compounds (Xia et al., 2018; Wu et al., 2018; Wan et al., 2019) could affect the hepatic lipid metabolism.

In this study, the serum metabolites of F1 offspring were determined by a NeoBase Nonderivatized MS/MS Kit and analyzed using a tandem mass spectrometry (MS/MS) system which is widely used for the screening of some genetic metabolic diseases (Chace et al., 1993; Chace et al., 2002; Rashed, 2001). There were totally 11 kinds and 15 kinds of metabolites changed significantly after 0.5 and 5 μ m maternal MPs treatment, respectively. For the female F1 offspring, most amino acids tended to decrease while most amino acids for the male F1 offspring tended to increase after maternal MPs treated, and reflected gender differences (Fig. 4A). Della et al. (2018) used the omics profiling of liver metabolism and proved a major sex difference that females maintain the synthesis of energy storage molecules (lipids) at the expense of amino acids, and males simply slow down anabolic pathways. The results demonstrated that female and male F1 offspring react differently to maternal MPs exposure during gestation while the specific mechanism deserves further study.

In addition to this, the free carnitine and acyl-carnitine were also changed, the contents of C0 increased significantly in the

serum of female offspring; while the long chain acylcarnitines (C16, C16:1, C18, C18:1) were decreased significantly for male offspring (Fig. 4B). The levels of C0 and C0/(C16 + C18) are important indicators for clinical screening of genetic diseases (Wilcken and Wiley, 2008; El-Gharbawy and Vockley, 2018), and the both indicators in our study tended to increase, which showed a close relationship to fatty acids metabolism disorder. Subsequently, the hepatic transcriptional levels of genes involving FA transportation, β -oxidation and FA synthesis were also verified the previous results and speculation. Through the results of gene expression, it was found that the gene expression involving β -oxidation (including *Ppar- α* , *Acox*, *Cpt α* and *Mcad*) were mostly suppressed after maternal 5 μ m MP treatment (Fig. 5D and E), which might result in energy supply problem of the body. Nevertheless, after a short-term exposure of ~45 nm nanoplastics (NPs) on the marine fish *Dicentrarchus labrax*, abundance of mRNA transcripts related to lipid metabolism, *Ppar- α* and *Ppar- γ* , were significantly increased (Brandts et al., 2018). In addition, the relative expression of *Ppar- α* and *Ppar- γ* were also altered after MPs exposure both in mice and zebrafish (Lu et al., 2018; Wan et al., 2019). The Peroxisome proliferator-activated receptors (PPARs) were found to be key regulators of lipid and carbohydrate metabolism and even able to modulate inflammatory responses, indicating that PPAR functions could also be studied in animal models of disease (Varga et al., 2011). These remind us that PPARs might be the entry point in studying the mechanisms of MPs induced metabolism disorder.

In this study, maternal MPs exposure induced the metabolic disorders in their offspring, even though the offspring were not directly exposed to MPs. From our point of view, there may be two reasons for the metabolic disorder of the offspring: (1) maternal metabolism was changed due to MPs exposure, thus induces intergenerational effects on the F1 offspring (prove as Fig. S4, many biochemical parameters and hepatic gene expression of F0 dams were altered after MPs exposure); (2) according to previous study, small sizes MPs could pass through placenta, and even transfer to the next generation (Bouwmeester et al., 2015). Both human and animal studies proved that environmental exposures during early life including in utero environment, maternal nutrition, stress, and exposure to chemicals (such as environmental pollutants), has long-term influence on susceptibility of offspring to chronic diseases at adulthood (Rosenfeld, 2012; Kaelin Jr. and McKnight, 2013; Sales et al., 2017). In addition, in recently years, the trans generational effects of MPs were carried out in some different models including oysters, *Daphnia magna*, zebrafish and *Caenorhabditis elegans* etc. (Sussarellu et al., 2016; Zhao et al., 2017; Martins and Guilhermino, 2018; Pitt et al., 2018). For example, Sussarellu et al. (2016) found that 2 months polystyrene MPs exposure caused significantly decrease in oocyte number, diameter and sperm velocity of exposed parents and the D-larval yield and larval development of offspring from exposed parents derived decreased by 41% and 18%, respectively. Similar inhibitions on growth, reproduction and population growth rate were observed in *Daphnia magna* model, and the results also indicated that recovery from chronic exposure to microplastics may take several generations (Kaelin Jr. and McKnight, 2013). In contrast, the effects of MPs exposure during gestation on the growth of offspring was limited in the current study. The transgenerational toxicity of MPs might be due to the translocation of them in to reproductive organs that potentially led to the transfer to the next generation. Interestingly, nanoplastics (NPs) were observed in the next generation of some lower trophic organisms such as zebrafish and *Caenorhabditis elegans* (Zhao et al., 2017; Pitt et al., 2018). Whether the MPs could transfer from the mother to the progeny through the placental barrier or not in animals still remained unclear, more future works should be carried out to find direct evidence.

In summary, maternal MPs exposure during gestation had no significant effect on growth of their F1 offspring while the changes of amino acids metabolism and carnitine metabolism homeostasis prompt increasing health risks relevant to fatty acids metabolism disorder. Moreover, the consequences induced by maternal MPs treatment seems to have a close relationship with particle size, metabolic changes (especially for relevant genes expression) caused by 5 μ m MP were more pronounced. We hope that these findings can provide new data support for the health risk assessment of MPs.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envpol.2019.113122>.

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